

A novel high-order linguistic time series forecasting model with the growth of declared word-set

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Abstract—The existing researches have shown that the fuzzy forecasting methods based on the high-order fuzzy time series are better than the fuzzy forecasting methods based on the first-order fuzzy time series. The linguistic forecasting method based on the first-order linguistic time series which can handle directly the word-set of the linguistic variable have been examined by Hieu et al. This paper examines a novel model of high-order linguistic time series with the growth of declared word-set. A procedure for forecasting the enrollments of University of Alabama and Lahi crop production of Indian is developed based on the proposed model. In the proposed forecasting method, the high-order linguistic logical relationship groups will be established and utilized to calculate the forecasted values based on the quantitative semantics of the used words generated by the hedge algebras structure. The experimental results show that the forecasting accuracy of the proposed high-order forecasting method is better than their counterparts and the growth of the word-set of the linguistic variable is significant in increasing the accuracy of the forecasted results.

Keywords—linguistic time series, high-order linguistic time series, linguistic logical relationship, hedge algebras, time series forecasting.

I. INTRODUCTION

The subject of the time series forecasting model has attracted substantial attention in recent years. For almost the realistic cases, the forecasting problem is uncertainty. In other words, it seems impossible to predict the future values of things and phenomena with one hundred percent of accuracy. However, we can examine different forecasting models and find the best fit model for a certain dataset.

The living of human is a sequence of the decision-making process. In reality, they always use their linguistic experience to make decisions or in other words, they save their experiences of the real world by linguistic words and use them to make decisions in the future. In computer science, researchers have found many ways to simulate the human behaviors into the machines. The most successful one is fuzzy set theory approach. Fuzzy set theory proposed by Zadeh [1] uses fuzzy sets to represent the computational semantics of linguistic words. The utilization of fuzzy set theory to time series forecasting problem first proposed by Song and Chissom in 1993 [2]–[4]. For example, we can use linguistic words such as “small”, “quite small”, “very small”, “large”, “rather large”, “extremely large”, etc. to describe the number of students who enroll to a university by years. By the fuzzy time series, the mentioned linguistic

words can be represented by fuzzy sets. There are many publications of the fuzzy forecasting methods based on the fuzzy time series which can be found in the literature.

Chen in [5] examined the high-order fuzzy time series forecasting model where the n th-order fuzzy logical relationship can be derived from the fuzzified historical data. In recent two decades, among so many improvements of the fuzzy time series forecasting methods, a number of high-order fuzzy time series forecasting methods has been proposed such as [6]–[23]. Among those studies, Singh [17] proposed a computational method of forecasting for the fuzzy time series model using high-order fuzzy logical relationships of differences of consecutive values. Most of the improvements in high-order fuzzy time in recent years are the proposal of hybrid forecasting models with modern computational techniques using artificial neural networks, machine learning, and evolutionary computation, etc.

Despite many important achievements, in the computation model of fuzzy time series, there is no formal connection between the fuzzy sets and the inherent semantics of the linguistic words represented by the fuzzy sets. In other words, the linguistic words are used in the fuzzy time series merely as the linguistic labels of fuzzy sets. Therefore, the relationships between the qualitative semantics of linguistic words and their fuzzy sets based semantics have not been thoroughly examined in the fuzzy time series forecasting models.

In the other side of natural language processing method, hedge algebras (HA) was introduced by Ho and Wechler in 1990 [24]. In hedge algebras, the linguistic variables can be modeled by an axiomatic way using inherent quantitative semantics of linguistic values. Many applications of hedge algebras can be found in the field of fuzzy control [25]–[31], data classification and regression using fuzzy rule-based systems [32], [33], image processing [34] and so on. In 2020, Hieu et al. [35], [36] proposed linguistic time series (LTS) forecasting model handling the linguistic words immediately based on the theory of hedge algebras. It is the first time that the constructed linguistic time series can simulate the way human users describe numerical time series in terms of their linguistic words.

In this paper, inspired by the mentioned researches of fuzzy time series, we propose a novel high-order linguistic time series forecasting model. In the proposed model, the current linguistic value of the linguistic variable X_L depend not only on the adjacent previous word but also the series of preceding words.

In addition, the forecasting linguistic variable $\mathcal{X}_L$ is considered as the linguistic counterpart of the numerical forecasting one $\mathcal{X}_N$. So, we assume that there is not any restriction on the declared word set's cardinality of $\mathcal{X}_L$. The accuracy of the proposed high-order LTS forecasting models can considerably increase when the declared word-set of $\mathcal{X}_L$ grows. The proposed linguistic forecasting method is applied to the problems of the enrollments of University of Alabama and the Lahi crop production of Indian to illustrate the effectiveness of the proposed forecasting model and the scalability of the word-set of $\mathcal{X}_L$. Besides, this study would like to present a comprehensive formalized methodology for handling linguistic words directly and simulate the way human users making a forecasting process in terms of their words. Therefore, we do not apply any optimization techniques which may improve the accuracy of forecasted results.

The rest of the paper will be organized as follows. Section II briefly introduces some fundamental knowledges of hedge algebras and the linguistic time series forecasting model. The proposed novel high-order linguistic time series forecasting model is also presented in this section. The experimental studies will be introduced and discussed in Section 3. Finally, the paper will be covered with the conclusions.

II. LINGUISTIC TIME SERIES AND ITS FORECASTING MODEL

A. Some basic knowledges of hedge algebras

In the human society, people utilize their historical experiences to communicate one to another by linguistic words in natural language. Linguistic words themselves reveal their inherent semantics. Because of their qualitative semantics, every word has its semantic order relation with each other in the word-domain of a linguistic variable. That motivated Ho and Wechler [24], [37] to propose hedge algebras which can deal with linguistic words directly.

Let $\mathcal{A}X = (X, G, C, H \leq)$ be a hedge algebras (HA), where X is the word-domain of a given linguistic variable $\mathcal{X}_L$; G is the set of two opposite generators, the negative one denoted by c^- and the positive one denoted by c^+ ; C is the set of three constants 0, W , 1 which are the smallest word, the neutral and the largest ones (e.g., such specific words are “*absolutely small*”, “*medium*” and “*absolutely large*”); H is the set of hedges such as “*little*”, “*rather*”, “*very*”, “*extremely*”, and so on; $\leq$ is the semantic order relation in the word-domain.

Each linguistic value in HA structure has its own semantic. In order to model the quantitative semantics of all words in the word-domain, it is necessary to construct two notions of HA which are semantically quantifying mapping (SQM) and fuzziness measure, denoted by $v : X \rightarrow [0, 1]$ and $m : X \rightarrow [0, 1]$, respectively. SQM and fuzziness measure of HA structure can be defined as follows.

Definition 1. [38] A function $m : X \rightarrow [0, 1]$ is said to be *fuzziness measure* of terms in X provided that:

- $m(c^-) + m(c^+) = 1$;
- $\sum_{h \in H} m(hu) = m(u)$, for $\forall u \in X$;
- $m(0) = m(W) = m(1) = 0$, fuzziness measure of all constants is 0;

For $\forall x, y \in X, \forall h \in H, \frac{m(hx)}{m(x)} = \frac{m(hy)}{m(y)}$, this proportion does not depend on specific elements and, hence, it is called fuzziness measure of the hedge h and denoted also by $\mu(h)$ for simplicity.

The fuzziness measure m of a linguistic variable $\mathcal{X}_L$ has the following properties:

- m1) $m(hx) = \mu(h)m(x)$, for $\forall x \in X$;
- m2) $m(c^-) + m(c^+) = 1$;
- m3) $\sum_{-q \leq i \leq p, i \neq 0} m(h_i x) = m(x)$, $\forall x \in X$;
- m4) Set $\sum_{-q \leq i \leq -1} \mu(h_i) = \alpha$, $\sum_{1 \leq i \leq p} \mu(h_i) = \beta$, where $\alpha, \beta > 0$ and $\alpha + \beta = 1$.

In the computational model of HA, each linguistic word or hedge has its sign defined as follows.

Definition 2. [38] A function $sign : X \rightarrow \{-1, 1\}$ is a mapping which is defined recursively as follows. For $h, h' \in H$ and $c \in \{c^-, c^+\}$:

- 1) $sign(c^-) = -1, sign(c^+) = +1$;
- 2) $sign(hc) = -sign(c)$ ($= +sign(c)$) if h is negative (positive) with respect to c ;
- 3) $sign(h'hx) = -sign(hx)$ if $h'hx \neq hx$ and h' is negative with respect to h ;
- 4) $sign(h'hx) = +sign(hx)$ if $h'hx \neq hx$ and h' is positive with respect to h .

Definition 3. [38] For given values of the fuzziness parameters of a variable $\mathcal{X}_L$, its corresponding SQM $v : X \rightarrow [0, 1]$ is defined as follows:

- 1) $v(W) = \theta = m(c^-)$;
- 2) $v(c^-) = \theta - \alpha m(c^-)$ and $v(c^+) = \theta + \alpha m(c^+)$;
- 3) $v(h_j x) = v(x) + sign(h_j x) \left\{ \sum_{i=sign(j)}^j m(h_i x) - \omega(h_j x) m(h_j x) \right\}$

where $\omega(h_j x) = \frac{1}{2} [1 + sign(h_j x) sign(h_p h_j x) (\beta - \alpha)] \in \{\alpha, \beta\}$.

SQM and fuzziness measure are two important notions of HA. More specifically, for given α and θ parameters, all the components of the HA's computational model can be obtained.

B. Linguistic time series (LTS)

Humans use their own language capacity to collect the data. Their brain saves all information in the real world by linguistic experience of specific owner. Human also use their own experience to make decision. In the time series forecasting problem, simulating the behavior of human's decision making process, the model also uses preceding data to predict the future value of time series data. When all the points of historical time series data are linguistic values, we have the LTS.

Definition 4. [35] Let $\mathbb{X}$ be a set of linguistic words in the natural language of a variable $\mathcal{X}$ defined on the universe of discourse U_X to describe its numeric quantities. Then, any series $L(t), t = 0, 1, 2, \dots$, where $L(t)$ is a finite collection of words of $\mathbb{X}$, is called a *linguistic time series*.

Definition 5. [35] Let $\mathcal{F}$ be a LTS. In $\mathcal{F}$, if every time $t = k - 1$, the linguistic value of LTS is x_i and x_j is the linguistic value of $\mathcal{F}$ at the time $t = k$, then we have *linguistic logical relationship* of the form $x_i \rightarrow x_j$.

Definition 6. [35] Suppose we have more than one linguistic logical relationship with the same left-hand side denoted by $x_i \rightarrow x_{j_1}, x_i \rightarrow x_{j_2}, \dots, x_i \rightarrow x_{j_n}$, then we can group them into the *linguistic logical relationship group* denoted by $x_i \rightarrow x_{j_1}, x_{j_2}, \dots, x_{j_n}$.

In Def. 5, the logical relationship $x_i \rightarrow x_j$ indicates that the linguistic value x_j to be caused by the only linguistic value x_i , called the first-order linguistic logical relationship. If there is more than one element on the left-hand side of the relationship, then we have the high-order linguistic logical relationship.

Definition 7. In the linguistic time series $\mathcal{F}$, if x_k , the linguistic value of $\mathcal{X}$ at the time k , to be caused by $x_{k-1}, x_{k-2}, \dots, x_{k-\lambda}$ are the linguistic values of $\mathcal{X}$ at the time $k - 1, k - 2, \dots, k - \lambda$, respectively, then the linguistic logical relationship is presented by:

$$x_{k-\lambda}, \dots, x_{k-2}, x_{k-1} \rightarrow x_k,$$

called the λ th-order linguistic logical relationship.

In a similar way to Def. 6, the high-order linguistic logical relationship group can be established based on individual high-order linguistic logical relationship, denoted by

$$x_{k-\lambda}, \dots, x_{k-2}, x_{k-1} \rightarrow x_{k_1}, x_{k_2}, \dots, x_{k_m},$$

where λ is the order of the linguistic logical relationship group.

C. High-order LTS forecasting model

Using the high-order linguistic logical relationship groups, the high-order LTS forecasting model was established and briefly reviewed as follows.

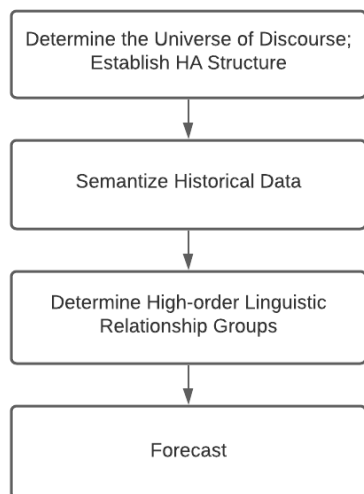


Fig. 1. The high-order linguistic time series forecasting model

In the first step, the universe of discourse and the HA structure with syntactic, qualitative and quantitative semantics are determined. The second step computes the numerical semantics of the declared words of $\mathcal{X}_L$. In the third step,

according to the historical data, the high-order linguistic logical relationship groups, used in the next step, were observed. Using the linguistic logical relationship groups, ones can calculate the forecasted values of the time series data. Some performance measures can be utilized to assert the effective of the forecasting model.

III. SOME EXPERIMENTAL RESULTS AND DISCUSSIONS

This section presents the experimental results of the proposed high-order LTS forecasting models and shows their performance in comparison with their counterpart models. The comparative study is designed to compare the proposed high-order LTS forecasting model performance with those examined by Chen [5] and Singh [17], using the University of Alabama historical data observed from 1971 to 1992, and Lahi crop product of Indian.

The experiments are executed to show the following statements:

- The proposed high-order LTS forecasting models outperform their counterpart of fuzzy approaches with restricted linguistic words.
- In the proposed high-order LTS forecasting models, the forecasting linguistic variable $\mathcal{X}_L$ is considered as the linguistic counterpart of the numerical forecasting one $\mathcal{X}_N$. So, we assume that there is not any restriction on its declared word set's cardinality. The accuracy of the proposed high-order LTS forecasting models can considerably increase when the declared word-set of $\mathcal{X}_L$ grows.

A. Forecast the enrollments of University of Alabama

1) Compare with the two counterparts examined by Chen and Singh

To show the efficiency of the proposed high-order LTS forecasting models, we apply them to the historical data of the enrollments of University of Alabama observed from 1971 to 1992, as shown the column "Actual Enrollment" in Tab. I. To be more convenient, the experiment follows the procedure as follows:

Step 1: Determine the syntactic, qualitative and quantitative semantics of the forecasting linguistic variable $\mathcal{X}_L$.

- The syntactic semantics are determined by declaring two primary words $c^- = \text{small}$, $c^+ = \text{large}$ (I) and the linguistic hedges. In this study, we just use two linguistic hedges $H^- = \{\text{Little } (L)\}$ and $H^+ = \{\text{Very } (V)\}$. For the model illustration and comparative study, the cardinality of the declared word-set of $\mathcal{X}_L$ is set to 9 (2-specificity level). So, we have: $U_{\mathcal{X}_L} = X_{(2)} = \{0, V_small, small, L_small, medium, L_large, large, V_large, 1\}$, where 0 stands for *Absolutely small* and 1 stands for *Absolutely large*.
- The qualitative semantics of $\mathcal{X}_L$ is specified by the signs of hedges and the relative signs between them: $\text{sign}(L) = \text{sign}(L, L) = \text{sign}(L, V) = -1$ and $\text{sign}(V) = \text{sign}(V, V) = \text{sign}(V, L) = +1$.
- The quantitative semantics is defined by specifying the fuzziness parameter values of $\mathcal{X}_L$. In this study, they are

specified by human's experts and we have $m(c^-) = 0.437$, $\mu(L) = 0.511$.

Step 2: Compute the numerical semantics of the declared words of $\mathcal{X}_L$.

Transform the numerical semantics (SQM-values) of the declared words in the normalized domain $[0, 1]$ into the real numerical semantics in the universe of discourse $U_x = [13000, 20000]$ of $\mathcal{X}_L$ by the following formula:

$$v_R(l_i) = U_{x_{\min}} + (U_{x_{\max}} - U_{x_{\min}}) \times v(l_i), \quad (4)$$

where $l_i \in X_{(2)}$, $v(l_i)$ is the numerical semantic of the word l_i , $U_{x_{\min}}$ is the lower bound and $U_{x_{\max}}$ is the upper bound of U_x . Specifically, with the set $U_{x,L}$, $m(c^-) = 0.437$ and $\mu(L) = 0.511$, we have the real numerical semantics of the declared words: $U_{x,R} = \{13000, 13731, 14496, 15295, 16059, 17044, 18073, 19058, 20000\}$.

Step 3: Generate LTS from the given numerical time series.

Based on the real numerical semantics, the historical data of the enrollments are transformed into the linguistic words selected from the declared word-set of $\mathcal{X}_L$ in such a way that each historical data value is assigned a linguistic word which has its real numerical semantics closest to that historical data value at that timestamp. For example, the enrollment value of 15460 is of the year of 1975 closest to the real numerical semantics of 15295 of the word *Little small* in $U_{x,L}$. Therefore, that enrollment value is assigned the word *Little small*. By doing the same way for the enrollment data of other years, we obtain the LTS as shown in the column “**Constructed LTS**” in *Tab. I*. Once the LTS is constructed, the high-order linguistic logical relationships (LLRs) and the high-order linguistic logical relationship groups (LLRGs) are created. Just for the illustration of the case of 2-specificity level of $\mathcal{X}_L$ (9 declared words), the second-order and third-order LLRGs are shown in *Tab. II* and *Tab. III*, respectively.

Step 4: Calculate the crisp forecasted values. Two applied computation rules are as follows:

- (1) If an high-order LLRG is of the form: $l_{i1}, l_{i2}, \dots, l_{ip} \rightarrow l_{j1}, l_{j2}, \dots, l_{jp}$ ($\lambda, p \geq 1$), where $l_{i1}, l_{i2}, \dots, l_{ip}$ is the current state of a year, say k and $v_R(l_{j1}), v_R(l_{j2}), \dots, v_R(l_{jp})$ are the real numerical semantics of the words $l_{j1}, l_{j2}, \dots, l_{jp}$, respectively, then the numeric forecasted value of year $k+1$ is $(v_R(l_{j1}) + v_R(l_{j2}) + \dots + v_R(l_{jp}))/p$.
- (2) If the current state of year k is $l_{i1}, l_{i2}, \dots, l_{ip}$ and there does not exist any word in the right-hand side of any high-order LLRG, then, the forecasted value of year $k+1$ is $(v_R(l_{i1}) + v_R(l_{i2}) + \dots + v_R(l_{ip}))/\lambda$.

TABLE I. THE LINGUISTIC TIME SERIES OF THE ENROLLMENTS OF THE UNIVERSITY OF ALABAMA FROM 1971 TO 1992 USING 9 WORDS.

Year	Actual Enrollment	Constructed LTS
1971	13055	<i>Absolutely small</i>
1972	13563	<i>Very small</i>

Year	Actual Enrollment	Constructed LTS
1973	13867	<i>Very small</i>
1974	14696	<i>small</i>
1975	15460	<i>Little small</i>
1976	15311	<i>Little small</i>
1977	15603	<i>Little small</i>
1978	15861	<i>medium</i>
1979	16807	<i>Little large</i>
1980	16919	<i>Little large</i>
1981	16388	<i>medium</i>
1982	15433	<i>Little small</i>
1983	15497	<i>Little small</i>
1984	15145	<i>Little small</i>
1985	15163	<i>Little small</i>
1986	15984	<i>medium</i>
1987	16859	<i>Little large</i>
1988	18150	<i>large</i>
1989	18970	<i>Very large</i>
1990	19328	<i>Very large</i>
1991	19337	<i>Very large</i>
1992	18876	<i>Very large</i>

TABLE II. THE SECOND-ORDER LLRGs USING 9 WORDS.

Group	2-order LLRG
1	$A_small, V_small \rightarrow V_small$
2	$V_small, V_small \rightarrow small$
3	$small, V_small \rightarrow L_small$
4	$L_small, small \rightarrow L_small$
5	$L_small, L_small \rightarrow L_small, medium$
6	$medium, L_small \rightarrow L_large$
7	$L_large, medium \rightarrow L_large, large$
8	$L_large, L_large \rightarrow medium$
9	$medium, L_large \rightarrow L_small$
10	$L_small, medium \rightarrow L_small$
11	$large, L_large \rightarrow V_large$
12	$V_large, large \rightarrow V_large$
13	$V_large, V_large \rightarrow V_large$

TABLE III. THE THIRD-ORDER LLRGs USING 9 WORDS

Group	3-order LLRG
1	$V_small, V_small, A_small \rightarrow small$
2	$small, V_small, V_small \rightarrow L_small$

Group	3-order LRRG
3	$L_small, small, V_small \rightarrow L_small$
4	$L_small, L_small, small \rightarrow L_small$
5	$L_small, L_small, L_small \rightarrow medium, L_small$
6	$medium, L_small, L_small \rightarrow L_large$
7	$L_large, medium, L_small \rightarrow L_large, large$
8	$L_large, L_large, medium \rightarrow medium$
9	$medium, L_large, L_large \rightarrow L_small$
10	$L_small, medium, L_large \rightarrow L_small$
11	$L_small, L_small, medium \rightarrow L_small$
12	$large, L_large, medium \rightarrow V_large$
13	$V_large, large, L_large \rightarrow V_large$
14	$V_large, V_large, large \rightarrow V_large$
15	$V_large, V_large, V_large \rightarrow V_large$

In this study, the mean square error (*MSE*) and average forecasting error (*AFE*) are used to evaluate the forecasting

models, where F_i and A_i are the forecasted value and the actual value, respectively; ϵ_i is forecasting error of year i :

$$MSE = \left(\frac{1}{N} \right) \sum_{i=1}^N (F_i - A_i)^2 \quad (5)$$

$$\epsilon_i \text{ (in percent)} = \frac{|F_i - A_i|}{A_i} \times 100 \quad (6)$$

$$AFE \text{ (in percent)} = \frac{\epsilon_i}{N} \quad (7)$$

The forecasted enrollments by the proposed high-order (from 2-order to 9-order) LTS forecasting models using 9 declared words are shown in *Tab. IV*. The comparison of MSE values with three other high-order fuzzy time series forecasting models are shown in *Tab. V*. The MSE values of the proposed high-order LTS forecasting models are much better than all their counterparts (Chen [5], Hwang et al. [39] and Singh [17]) from the 4-order to 9-order. Evaluated based on AFE values, it is easy to see in *Tab. VI* that all proposed high-order LTS forecasting models have AFE values smaller than their counterparts (Song and Chissom [4], Hwang et al. [39] and Singh [17]).

TABLE IV. FORECASTED ENROLLMENTS BY PROPOSED HIGH-ORDER LTS FORECASTING MODELS USING 9 AND 17 DECLARED WORDS.

Year	Actu. Enrol.	Proposed (9 declared words)								Proposed (17 declared words)							
		2-order	3-order	4-order	5-order	6-order	7-order	8-order	9-order	2-order	3-order	4-order	5-order	6-order	7-order	8-order	9-order
1971	13055																
1972	13563																
1973	13867	13731								13750							
1974	14696	14496	14496							14644	14644						
1975	15460	15295	15295	15295						15537	15537	15537					
1976	15311	15295	15295	15295	15295					15169	15169	15169	15169				
1977	15603	15677	15295	15295	15295	15295				15353	15537	15537	15537	15537			
1978	15861	15677	15677	16059	16059	16059	16059			15795	15795	15795	15795	15795	15795		
1979	16807	17044	17044	17044	17044	17044	17044	17044		16689	16689	16689	16689	16689	16689	16689	
1980	16919	17558	17558	17558	17558	17044	17044	17044	17044	16689	16689	16689	16689	16689	16689	16689	16689
1981	16388	16059	16059	16059	16059	16059	16059	16059	16059	16164	16164	16164	16164	16164	16164	16164	16164
1982	15433	15295	15295	15295	15295	15295	15295	15295	15295	15537	15537	15537	15537	15537	15537	15537	15537
1983	15497	15295	15295	15295	15295	15295	15295	15295	15295	15537	15537	15537	15537	15537	15537	15537	15537
1984	15145	15677	15295	15295	15295	15295	15295	15295	15295	15169	15169	15169	15169	15169	15169	15169	15169
1985	15163	15677	15677	15295	15295	15295	15295	15295	15295	15353	15169	15169	15169	15169	15169	15169	15169
1986	15984	15677	15677	16059	16059	16059	16059	16059	16059	16164	16164	16164	16164	16164	16164	16164	16164
1987	16859	17044	17044	17044	17044	17044	17044	17044	17044	16689	16689	16689	16689	16689	16689	16689	16689
1988	18150	17558	17558	17558	17558	18073	18073	18073	18073	18053	18053	18053	18053	18053	18053	18053	18053
1989	18970	19058	19058	19058	19058	19058	19058	19058	19058	18855	18855	18855	18855	18855	18855	18855	18855
1990	19328	19058	19058	19058	19058	19058	19058	19058	19058	19327	19327	19327	19327	19327	19327	19327	19327
1991	19337	19058	19058	19058	19058	19058	19058	19058	19058	19327	19327	19327	19327	19327	19327	19327	19327

Year	Actu. Enrol.	Proposed (9 declared words)								Proposed (17 declared words)							
		2-order	3-order	4-order	5-order	6-order	7-order	8-order	9-order	2-order	3-order	4-order	5-order	6-order	7-order	8-order	9-order
1992	18876	19058	19058	19058	19058	19058	19058	19058	19058	18855	18855	18855	18855	18855	18855	18855	18855
MSE		98783	94001	78665	81691	40704	37093	36943	35464	17996	13264	13851	14313	13951	14591	15322	15429
AFE		1.60	1.58	1.37	1.39	1.11	1.05	1.03	1.0	0.69	0.56	0.57	0.57	0.55	0.56	0.57	0.56

TABLE V. COMPARISON OF MSE VALUES.

Methods/MSE	2-order	3-order	4-order	5-order	6-order	7-order	8-order	9-order
Chen [5]	89,093	86,694	89,376	94,539	98,215	104,056	102,179	102,789
Hwang et al. [39]	333,171	299,634	315,489	278,919	296,950	316,720	301,228	306,485
Singh [17]	119,189	97,180	126,676	113,421	163,137	148,618	169,149	123,964
Proposed (9 declared words)	98,783	94,001	78,665	81,691	40,704	37,093	36,943	35,464

TABLE VI. COMPARISON OF AVERAGE ERROR (AFE) OF FORECASTS.

Methods/MSE	2-order	3-order	4-order	5-order	6-order	7-order	8-order	9-order
Song and Chissom [4]	3.15	3.89	4.37	4.41	4.49	4.35	4.45	4.23
Hwang et al. [39]	2.99	2.94	3.12	2.92	3.01	3.08	2.89	2.79
Singh [17]	1.80	1.56	1.74	1.68	2.07	1.89	1.98	1.65
Proposed (9 declared words)	1.60	1.58	1.37	1.39	1.11	1.05	1.03	1.0

2) The accuracy of the proposed high-order LTS forecasting models increases by allowing the current used word-set of the forecasting linguistic variable to grow

A special feature of the LTS forecasting models is that it considers the forecasting linguistic variable $\mathcal{X}_L$ as the linguistic counterpart of the numerical forecasting one. So, there is not any restriction on its declared word set's cardinality. This experimental study will show that the forecasting accuracy will considerably increase when the declared word-set of $\mathcal{X}_L$ grows. We assume that the cardinality of the declared word-set of $\mathcal{X}_L$ grows from 9 to 17, 33 and 65 words respectively corresponding to 3-, 4- and 5-specificity levels. For each declared word-set cardinality, the fuzziness parameter values of $\mathcal{X}_L$ should be adjusted suitably by human's experts as shown in Tab. VII. For example, in case of 17 declared words, the value of $m(c)$ is 0.527 and the values of $\mu(L)$ is 0.412, etc.

TABLE VII. FUZZINESS PARAMETER VALUES.

17 words		33 words		65 words	
$m(c)$	$\mu(L)$	$m(c)$	$\mu(L)$	$m(c)$	$\mu(L)$
0.527	0.412	0.65	0.35	0.65	0.35

Intuitively seen in Tab. IV and Tab. VIII, with the same order, the greater used linguistic words, the smaller corresponding MSE and AFE values we will have. Specifically, the 9-order LTS forecasting model using 65 words has its MSE value of 283 and its AFE value of 0.07, better than the respective ones of the other models with same order using 33, 17 and 9 words in turn, respectively, 1023 and 1.4, 15429 and 0.6, 35464 and 1.0. We also have the same analysis results for other order (from 2 to 8) using the same words. These comparison results state that the forecasting accuracy will considerably increase when the declared word-set of $\mathcal{X}_L$ grows and show the effectiveness of the proposed linguistic forecasting method.

TABLE VIII. FORECASTED ENROLLMENTS BY PROPOSED HIGH-ORDER LTS FORECASTING MODELS USING 33 AND 65 DECLARED WORDS.

Year	Actu. Enrol.	Proposed (33 declared words)								Proposed (65 declared words)							
		2-order	3-order	4-order	5-order	6-order	7-order	8-order	9-order	2-order	3-order	4-order	5-order	6-order	7-order	8-order	9-order
1971	13055																
1972	13563																
1973	13867	13812								13812							
1974	14696	14485	14485							14638	14638						
1975	15460	15520	15520	15520						15438	15438	15438					
1976	15311	15285	15285	15285	15285					15285	15285	15285	15285				

Year	Actu. Enrol.	Proposed (33 declared words)								Proposed (65 declared words)							
		2-order	3-order	4-order	5-order	6-order	7-order	8-order	9-order	2-order	3-order	4-order	5-order	6-order	7-order	8-order	9-order
1977	15603	15520	15520	15520	15520	15520				15673	15673	15673	15673	15673			
1978	15861	15958	15958	15958	15958	15958	15958			15958	15958	15958	15958	15958	15958		
1979	16807	16750	16750	16750	16750	16750	16750	16750		16795	16795	16795	16795	16795	16795	16795	
1980	16919	16877	16877	16877	16877	16877	16877	16877	16877	16877	16877	16877	16877	16877	16877	16877	16877
1981	16388	16388	16388	16388	16388	16388	16388	16388	16388	16388	16388	16388	16388	16388	16388	16388	16388
1982	15433	15520	15520	15520	15520	15520	15520	15520	15520	15438	15438	15438	15438	15438	15438	15438	15438
1983	15497	15520	15520	15520	15520	15520	15520	15520	15520	15520	15520	15520	15520	15520	15520	15520	15520
1984	15145	15158	15158	15158	15158	15158	15158	15158	15158	15158	15158	15158	15158	15158	15158	15158	15158
1985	15163	15158	15158	15158	15158	15158	15158	15158	15158	15158	15158	15158	15158	15158	15158	15158	15158
1986	15984	15958	15958	15958	15958	15958	15958	15958	15958	15958	15958	15958	15958	15958	15958	15958	15958
1987	16859	16877	16877	16877	16877	16877	16877	16877	16877	16877	16877	16877	16877	16877	16877	16877	16877
1988	18150	18176	18176	18176	18176	18176	18176	18176	18176	18152	18152	18152	18152	18152	18152	18152	18152
1989	18970	18965	18965	18965	18965	18965	18965	18965	18965	18965	18965	18965	18965	18965	18965	18965	18965
1990	19328	19327	19327	19327	19327	19327	19327	19327	19327	19327	19327	19327	19327	19327	19327	19327	19327
1991	19337	19327	19327	19327	19327	19327	19327	19327	19327	19327	19327	19327	19327	19327	19327	19327	19327
1992	18876	18838	18838	18838	18838	18838	18838	18838	18838	18882	18882	18882	18882	18882	18882	18882	18882
MSE		4234	4297	2062	1972	2053	1731	1182	1023	1284	1192	1078	1106	1133	882	273	283
AFE		0.28	0.28	0.21	0.20	0.20	0.18	0.15	0.14	0.16	0.15	0.13	0.13	0.13	0.11	0.07	0.07

B. Forecast the Lahi crop production of Indian

To prove that the proposed high-order LTS forecasting models are much more efficient than the ones proposed by Singh in [8], the proposed models are applied to forecast the Lahi crop production of Indian. With the observed production values are from 440 and 1067, we can set the value interval of the universe of discourse of the variable X_L to be $U_X = [400, 1100]$ as in [80]. Two primary words are $c^- = \text{low}$ (l) and $c^+ = \text{high}$ (h) and two linguistic hedges are *Little* (L) $\in H^-$ and *Very* (V) $\in H^+$. For illustration, the declared word-set of X_L in case of 2-specificity level is $U_{X,L} = X_{(2)} = \{0, \text{Very low}, \text{low}, \text{Little low}, \text{medium}, \text{Little high}, \text{high}, \text{Very high}, 1\}$. The fuzziness parameter values are chosen by human's experts as $m(c^-) = m(\text{low}) = 0.5$ and $\mu(L) = 0.51$ for all various orders and declared words. So, the real numerical semantics of the declared words

are $U_{X,R} = \{400, 484, 571.5, 662.5, 750, 837.5, 928.5, 1016, 1100\}$. Based on the given data setting, transform the observed historical data of the Lahi crop production shown the column "Actual production" in Tab. IX to the linguistic time series, establish the high-order LLRs and the high-order LLRGs, finally calculate the crisp forecasted values. The crop production forecasted results of the various high-order (from 2 to 9) LTS forecasting models are shown Tab. IX.

The comparisons of MSE and AFE values between the proposed LTS forecasting models with various orders using 9 words and the corresponding ones proposed by Singh in [8] are shown in Tab. X. Indeed, the comparison results prove that the proposed high-order LTS forecasting models are much better than the ones proposed by Singh on both evaluation methods of MSE and AFE.

TABLE IX. FORECASTED PRODUCTIONS BY PROPOSED HIGH-ORDER LTS FORECASTING MODELS USING 9 DECLARED WORDS.

Year	Actual production	Linguistic time series	2-order	3-order	4-order	5-order	6-order	7-order	8-order	9-order
1981	1025	V_high								
1982	512	V_low								
1983	1005	V_high	1016							
1984	852	L_high	837	837						
1985	440	A_low	400	400	400					
1986	502	V_low	484	484	484	484				

1987	775	Medium	750	750	750	750	750			
1988	465	V_low	484	484	484	484	484	484		
1989	795	L_high	837	837	837	837	837	837	837	
1990	970	high	929	929	929	929	929	929	929	929
1991	742	medium	750	750	750	750	750	750	750	750
1992	635	L_low	663	663	663	663	663	663	663	663
1993	994	V_high	1016	1016	1016	1016	1016	1016	1016	1016
1994	759	medium	750	750	750	750	750	750	750	750
1995	883	high	929	929	929	929	929	929	929	929
1996	599	low	572	572	572	572	572	572	572	572
1997	499	V_low	484	484	484	484	484	484	484	484
1998	590	low	572	572	572	572	572	572	572	572
1999	911	high	929	929	929	929	929	929	929	929
2000	862	L_high	837	837	837	837	837	837	837	837
2001	801	L_high	837	837	837	837	837	837	837	837
2002	1067	A_high	1100	1100	1100	1100	1100	1100	1100	1100
2003	917	high	929	929	929	929	929	929	929	929
MSE			713.6	743.3	770.5	724.4	748	755.7	782	711.9
AFE			3.3707	3.4846	3.5753	3.2689	3.2502	3.2518	3.1961	3.0471

TABLE X. COMPARISON OF MSE AND AFE VALUES.

Evaluation method	Forecasting models	2-order	3-order	4-order	5-order	6-order	7-order	8-order	9-order
MSE	Singh [17]	1089.7	908.9	1191.5	1311.6	1597.0	1207.2	1259.9	1425.8
	Proposed	713.6	743.3	770.5	724.4	748	755.7	782	711.9
AFE	Singh [17]	4.1359	3.8137	4.3566	4.4677	4.8753	3.9747	4.1337	4.5543
	Proposed	3.3707	3.4846	3.5753	3.2689	3.2502	3.2518	3.1961	3.0471

IV. CONCLUSIONS

The aims of the study in this paper is to establish a formalized methodology for handling words directly to solve the time series forecasting problems in such a way that it can simulate the way human users making a forecasting process in terms of their words. In fact, when observing the numerical time series data, human users can quickly transform its variations into the word variations which form a linguistic time series. They can do reasoning in their mind in some ways based on the relationships of the words in linguistic time series to estimate the forecasting values. This paper focuses on examining the high-order linguistic time series to establish an efficient linguistic forecasting method in such a way that the declared word-set of the linguistic variable can grow to increase the forecasted results. The proposed forecasting method is applied to the forecasting problems of the enrollments of University of Alabama and Lahi crop production of Indian. The experimental results have shown that the proposed forecasting method is better than its counterparts and the growth of the declared word-set of the linguistic variable is significant in increasing the

accuracy of forecasted results of the linguistic forecasting method.

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